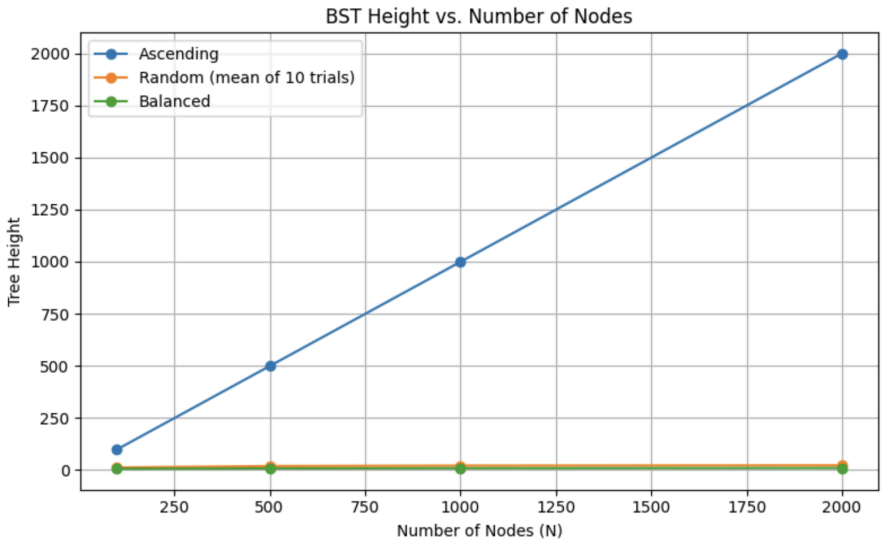
BST Report

AI Report:

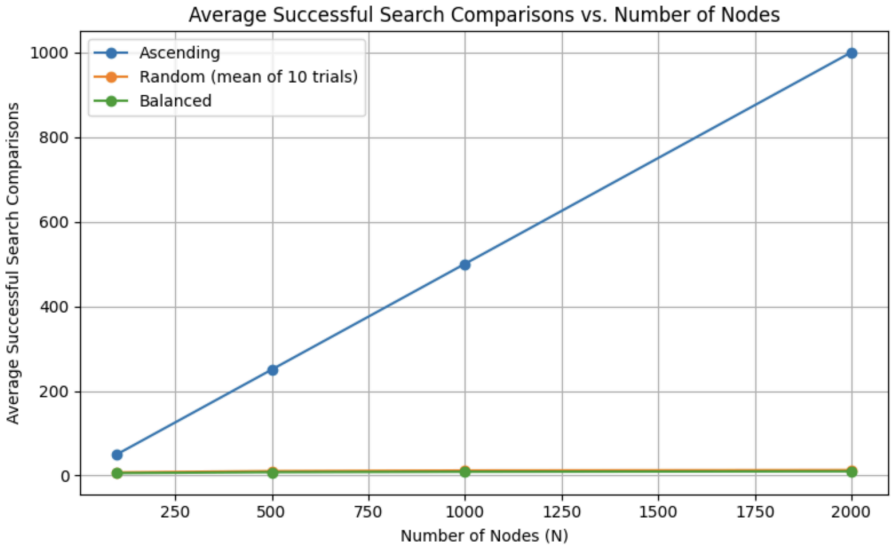
My first prompt informed Chat that I wanted it to help assist me by building phases that implement the key aspects of BST and how it can be used. Along the way it even created small quizzes that test my understanding of what we were doing and why. This is the entirety of my AI history, I apologize it is so long, I should have utilized it in a more efficient way when asking questions.
https://chatgpt.com/share/6ad6cb1d-13a0-83e8-bf9f-a5a01c3e1dc9

Analysis:

Following the implementation of these three strategies, the following can be concluded:



The tallest BST strategy came from the ascending insertion method. With the ascending order creating the worst *Big O* scenario, the height of the tree could be calculated as N-1. This is due to the creation of a linked-list or every additional element inserted being added to a single side of the tree. This is seen at a broader scoop when comparing the three strategies; balanced insertion having the minimum height implemented by its altering sequence and random insertion being somewhere between balanced and ascending, following the average case of *O(log(n))*



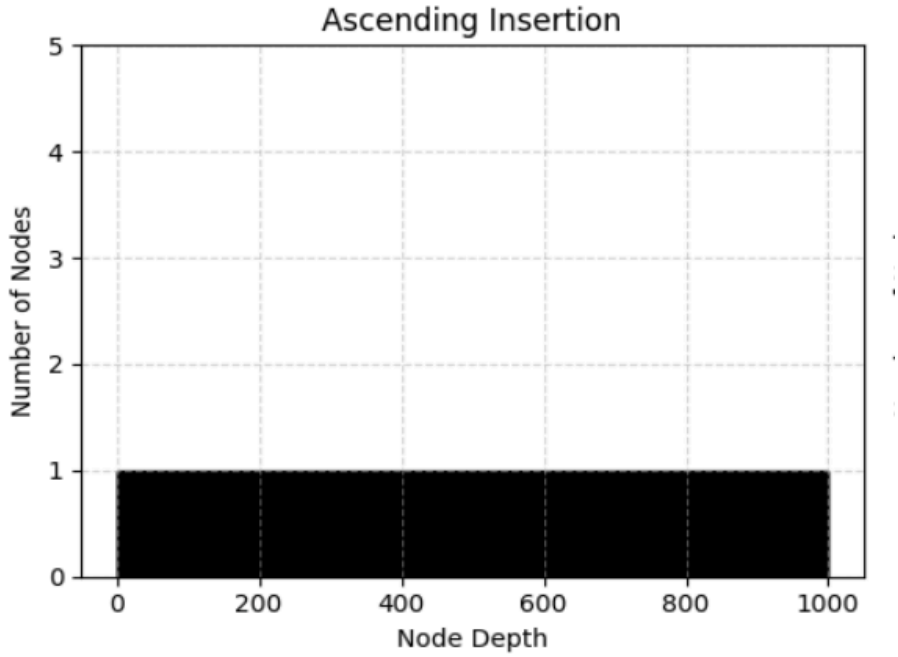
Similarly, the Average Successful Search Comparison (ASSC) follows closely to the nature of the tree's height. Ascending insertion will have an average search of the length of the list (or height / 2), whereas the balanced case should converge to the height itself and random insertion following slightly less closely to its height. Although random insertion can be balanced based on some probability, in most cases, it will not be balanced and it can be seen in the experiential results. Using $N = 100$, just two random cases alone, we can see that the heights ($height_1 = 14$, $height_2 = 20$) differ greatly due to the randomness that occurs.

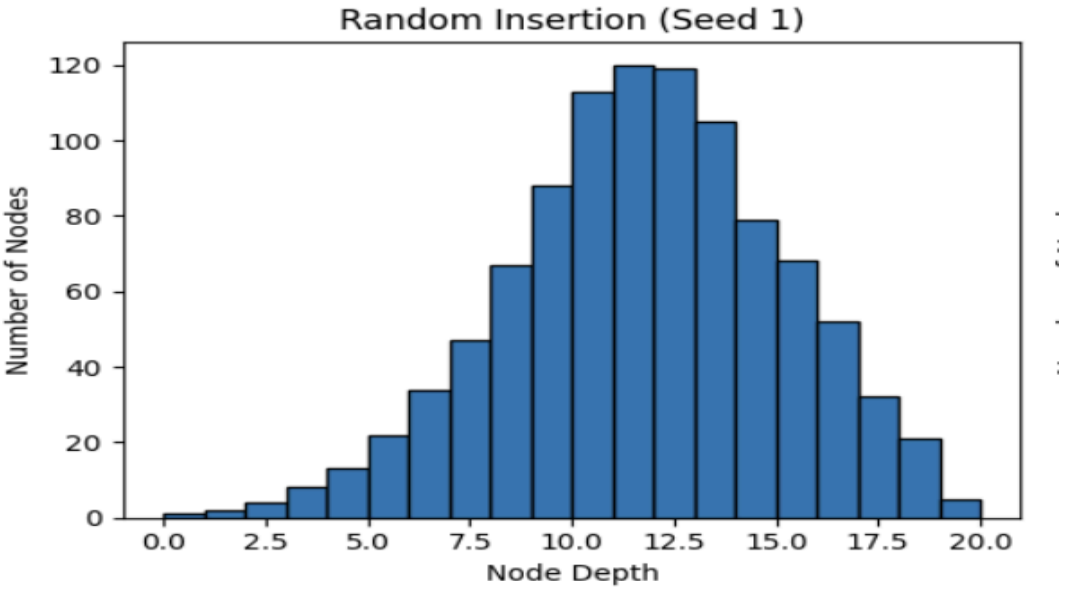
Node-depth distributions consist of ascending and random insertion following about half their heights, while balanced insertion follows closely to the height of the entire tree. Random insertion follows a unique random normal distribution whose probabilities remain within its max and min, which is expected. This adds additional variance among things including the ASSC. Even if two trees have similar or the same heights based on the computers need to check each possible case, therefore ASSC can be different due to the variance by insertion method.

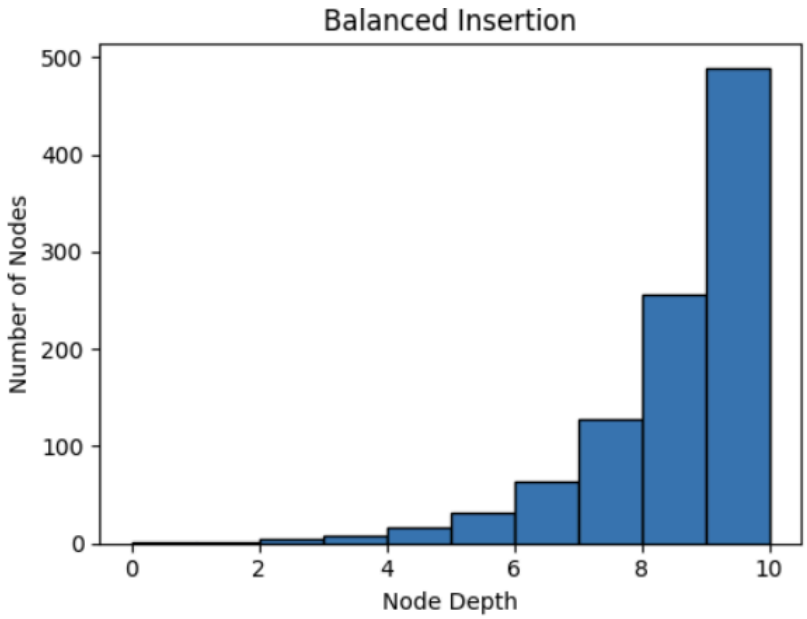
BST with a distinct key N, and assuming each key is searched exactly once, we can represent the relationship between average node depth and ASSC as:

$$Average\ node\ depth = ASSC - 1$$

This is due to the relationship found when the search compares the key at every node along the path, including the targeted node. Therefore, the average number of successful search comparisons is exactly one greater than the average node depth.







N	Strategy	Tree Height	Avg. Node Depth	Avg. Search Comparisons
100	Ascending	99.0	49.50	50.50
100	Random	12.3	6.57	7.57
100	Balanced	6.0	4.80	5.80
500	Ascending	499.0	249.50	250.50
500	Random	19.3	9.66	10.66
500	Balanced	8.0	7.00	8.00
1000	Ascending	999.0	499.50	500.50
1000	Random	21.4	11.10	12.10
1000	Balanced	9.0	7.99	8.99
2000	Ascending	1999.0	999.50	1000.50
2000	Random	22.9	12.02	13.02
2000	Balanced	10.0	8.98	9.98